RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532

CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENT

GROUP 6 – SOCIAL MEDIA ACCOUNT HACKING

August 2026

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CASE STUDY OBJECTIVES

- Understand the nature and impact of the suspected social-media account compromise.
- Identify the systems, accounts, devices and records that may contain digital evidence.
- Apply the six phases of the Digital Forensics Investigation Process to the case.
- Reconstruct the incident timeline using correlated digital artifacts.
- Prepare an evidence-based forensic report and recommend measures to prevent recurrence.

CASE STUDY SUMMARY

REC's website and official social media account were reportedly hacked. Unauthorized posts were uploaded from the account, and students and stakeholders received suspicious messages asking for money. Parents who became aware of these messages reported the incident to REC management. Management then asked the principal to conduct a Digital Forensic Investigation.

The investigation is treated as a suspected compromise of an institutional digital account. The forensic team must determine what happened, which systems and accounts were involved, what evidence exists, when the unauthorized activity occurred, and whether the available evidence can connect the activity to a particular device, account or network source.

INCIDENT IMPACT

Area	Possible Impact in the Case
Students & stakeholders	Exposure to suspicious money-request messages and possible financial/social-engineering risk.
Parents	Loss of trust and concern after receiving or learning about fraudulent communication.
Institutional reputation	Unauthorized posts from an official account can damage credibility and public trust.
Website / digital presence	Unauthorized access may affect official content, administrative controls or connected services.
Evidence & investigation	Deleting posts, logs or messages prematurely could make reconstruction more difficult.

INVESTIGATION QUESTIONS

- How was the official account accessed without authorization?
- When did the suspicious access begin and when were the unauthorized posts/messages created?
- Which devices, accounts, IP addresses or sessions were associated with the activity?
- Were account settings, passwords, recovery details or administrator privileges changed?
- What evidence supports the sequence of events and the final findings?

PART A – UNDERSTANDING THE CASE

1. WHAT IS THE CYBER INCIDENT DESCRIBED IN THIS CASE?

The cyber incident is the unauthorized compromise of REC's website and social media account, followed by the publication of unauthorized posts and the sending of suspicious messages requesting money. This indicates that an unauthorized party obtained access to, or control over, an institutional digital account.

2. WHAT TYPE OF DIGITAL CRIME IS SUSPECTED?

The suspected digital crime is **unauthorized access and account compromise**. The incident may also involve impersonation and attempted online fraud because the compromised institutional account was used to send money-request messages. The exact technique used to gain access must be determined from evidence rather than assumed.

3. WHO REPORTED THE INCIDENT?

The incident was reported to REC management by the **parents** after suspicious money-request messages became known to them. REC management then brought the incident to the principal, who initiated the digital forensic investigation.

4. WHAT ARE THE POSSIBLE SOURCES OF DIGITAL EVIDENCE?

Evidence Source	What It May Contain	Why It Matters
Social media account	Logins, sessions, posts, messages, security alerts	Shows unauthorized activity and access timing
REC website / server	Access and administrative logs	May reveal suspicious web activity
Admin computers / phones	Browser history, sessions, files, system artifacts	May reveal how the account was accessed
Institutional email	Security alerts and recovery messages	May show password-reset or new-login activity
Network devices	Firewall, VPN, proxy, DNS and connection records	Can correlate suspicious access with network activity
Screenshots / reports	Copies of posts and messages	Preserves visible evidence and impact
CCTV	Relevant physical activity	May correlate people with devices
USB / external media	Device identifiers and contents	Relevant if removable media is suspected

PART B – DIGITAL FORENSICS PROCESS

STEP 1 – IDENTIFICATION

WHAT HAPPENED?

REC management received reports that the official digital presence had been misused. Unauthorized posts appeared on the social media account and suspicious messages asking for money were received by students and stakeholders. These facts define the initial incident scope and justify forensic examination.

WHAT SYSTEMS ARE INVOLVED?

- REC's official social media account and its administrator/security settings.
- REC website, content-management system and relevant web servers if affected.
- Computers and mobile phones used by authorized account administrators.
- Institutional email accounts used for account recovery or security alerts.
- Network infrastructure such as firewall, VPN, proxy and router systems where records exist.
- Connected applications or services authorized to access the institutional account.

WHAT EVIDENCE MAY EXIST?

- Successful and failed social-media login records.
- IP addresses, device types, session information and login timestamps.
- Unauthorized posts, direct messages and attached links/files.
- Password reset, recovery and new-device notifications.
- Changes to password, recovery email/phone, administrator roles or connected applications.
- Website and server logs showing administrative or suspicious requests.
- Browser, email and device artifacts from authorized administrators.
- Screenshots and reports supplied by students, parents and stakeholders.

INITIAL INVESTIGATION CHECKLIST

Question	Initial forensic action
What was compromised?	Identify exact account(s), website components and administrative roles.
When was it noticed?	Record the earliest reliable report and detection time.
What changed?	Compare normal account settings/content with the unauthorized state.
What evidence may disappear?	Preserve logs, messages, sessions and security alerts immediately.
Who had legitimate access?	Create a list of authorized administrators and normal devices.

STEP 2 – PRESERVATION

WHY IT MATTERS

Digital evidence can be altered, deleted or overwritten easily. Preservation protects the original state of evidence so that later examination can be performed on reliable copies. In this case, posts, messages, account logs, security alerts and device artifacts may be especially fragile

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HOW WILL YOU PROTECT THE EVIDENCE?

- Secure the compromised account to prevent further misuse while preserving the relevant evidence.
- Do not delete suspicious posts, messages or account notifications before documenting them.
- Capture and preserve screenshots, URLs, timestamps and available platform security records.
- Preserve server, application, firewall and authentication logs before their normal retention period expires.
- Create forensic images or approved forensic extractions of relevant computers and mobile devices.
- Use write-blocking methods when acquiring storage media where applicable.
- Calculate SHA-256 or another approved cryptographic hash for forensic copies and record the values.

WHAT PRECAUTIONS SHOULD BE TAKEN?

- Only authorized investigators should handle original evidence.
- Every evidence transfer or access should be documented.
- Original evidence should be stored separately from working copies.
- Investigators should avoid unnecessary logins, reboots or changes to potentially relevant devices.
- The evidence collection method, date, time, source and responsible person should be recorded.
- Confidential account information should be protected from unnecessary disclosure.

WHY IS PRESERVING EVIDENCE IMPORTANT?

Preservation maintains evidence integrity and supports a defensible chain of custody. It also helps prevent the accidental loss of timestamps, sessions, logs and messages that may be required to reconstruct the incident.

PRESERVATION RECORD

Evidence	Immediate Preservation Action
Social-media posts/messages	Capture screenshots/export records and preserve timestamps before removal.
Login/session logs	Export or preserve platform security records and record the collection time.
Website/server logs	Copy relevant logs to secure forensic storage and preserve originals.
Admin device	Isolate as appropriate and acquire forensic image/extraction without altering original data.
Email security records	Preserve relevant alerts, password-reset messages and recovery notifications.

STEP 3 – COLLECTION

After preservation, the forensic team collects relevant evidence systematically. The objective is to obtain information that can establish the access path, incident timeline, affected accounts, associated devices and possible network sources.

Evidence	Collection Method / Content	Forensic Value
Social-media account	Login/session history, posts, messages, security settings and alerts	Shows access and unauthorized activity
Website/server	Authentication, access and administrative logs	Helps identify suspicious requests and changes
Computer / phone	Forensic image or approved extraction, browser and application artifacts	May reveal sessions, phishing or account activity
Email account	Security notifications, reset requests, recovery changes	May identify takeover indicators
Network logs	Firewall, VPN, proxy, DNS and connection records	Correlates IP/network activity
Screenshots	Unauthorized posts and money-request messages	Preserves visible content and impact
CCTV	Relevant time window around physical access	Correlates people and devices
USB / external devices	Device identifiers and relevant contents	Checks removable-media involvement

COLLECTION PRINCIPLES

- Collect the minimum relevant evidence needed to answer the investigation questions.
- Maintain evidence integrity and record acquisition details.
- Use forensic copies for examination rather than changing the original evidence.
- Assign unique evidence IDs and maintain chain-of-custody records.
- Record hashes for acquired forensic images or exported evidence where technically appropriate.

Example evidence identifiers: **E-01** Social-media account records; **E-02** unauthorized posts/messages; **E-03** security email records; **E-04** administrator device image; **E-05** server logs; **E-06** network logs; **E-07** CCTV.

STEP 4 – EXAMINATION

The Examination phase processes the collected evidence to identify useful artifacts. The goal is not simply to find suspicious information, but to extract information that can be correlated during the Analysis phase.

Examination Area	What Investigators Examine	Expected Information
Login history	Successful/failed logins, timestamps, IPs, devices	Normal vs. unusual access
Active sessions	Session/device list and session start times	Unknown or unexpected sessions
Account settings	Password, recovery email/phone, roles, connected apps	Unauthorized configuration changes
Posts & messages	Creation time, content, recipients, links, attachments	Sequence and impact of misuse
Browser history	Visited sites, downloads, phishing pages	Possible credential theft indicators
Email records	Security alerts and reset notifications	Evidence of takeover attempts
System artifacts	Event logs, application artifacts, file timestamps	Local activity around incident time
Network records	IP/connection events and security logs	Correlation with suspicious access

EXAMINATION OUTPUT

- An organized set of relevant account and device artifacts.
- A list of suspicious logins, sessions and configuration changes.
- A collection of timestamps for posts, messages, alerts and related events.
- Extracted network indicators that can be compared with account activity.
- An examination log documenting what evidence was reviewed and what was found.

STEP 5 – ANALYSIS

A. HOW WOULD YOU DETERMINE WHO ACCESSED THE ACCOUNT?

The team should correlate the social-media login history with device information, IP addresses, security alerts, administrator activity and local device artifacts. An unfamiliar device or IP that appears shortly before the unauthorized posts becomes an important investigative lead. However, the presence of an IP address or a legitimate user account does not by itself prove the identity of the attacker.

- List all authorized administrators and their normal devices.
- Identify unusual login times, locations, devices or sessions.
- Compare suspicious logins with the timestamps of unauthorized posts and messages.
- Check for password, recovery or administrator-role changes.
- Correlate platform evidence with email, device and network records.
- Use multiple independent evidence sources before reaching an attribution conclusion.

B. HOW WOULD YOU DETERMINE WHEN THE INCIDENT OCCURRED?

Investigators reconstruct a timeline from the last known legitimate activity, first suspicious login, security alerts, account-setting changes, unauthorized posts/messages, detection by parents or management, and the subsequent response. Timestamps should be normalized to a common time zone before comparison.

Stage	Possible Evidence	Purpose
Normal activity	Last known legitimate login/post	Establish baseline
Possible compromise	Unusual login / alert / password change	Locate probable access point
Unauthorized use	Posts, messages, setting changes	Establish misuse period
Detection	Parent report / management notification	Establish discovery
Response	Preservation and account-security actions	Record containment and evidence handling

C. WHETHER A USB DEVICE WAS USED?

USB involvement should only be considered if relevant computers or mobile devices contain evidence of removable-media use. Investigators can examine USB connection history, device identifiers, file-transfer artifacts and CCTV. A USB connection alone does not establish that it caused the account compromise; it must be correlated with other evidence.

STEP 6 – REPORTING

INCIDENT SUMMARY

REC's website and official social media account were reportedly compromised. Unauthorized posts were uploaded and suspicious messages asking for money were sent to students and stakeholders. Parents reported the issue to REC management, and the principal initiated a digital forensic investigation.

EVIDENCE COLLECTED

- Social-media login/session and security records.
- Copies or screenshots of unauthorized posts and suspicious messages.
- Website/server and application logs where available.
- Relevant administrator computer/mobile-device forensic artifacts.
- Email security and account-recovery records.
- Firewall, VPN, proxy and other network logs where available.
- CCTV footage relevant to physical access.
- USB/removable-device records where relevant.

FINDINGS

Finding	Interpretation
Unauthorized activity occurred	Posts and money-request messages were made from the institutional account without authorization.
Account access is central evidence	Login/session records and security alerts are key to reconstructing access.
Timeline can be reconstructed	Correlating timestamps can establish the sequence from suspicious access to misuse.
Device/network evidence may provide leads	IP and device records can support correlation with the suspicious sessions.
Attribution requires corroboration	No single IP, account or device artifact proves the identity of the attacker.

CONCLUSION

The investigation provides a structured evidence-based method for determining what happened, when the compromise occurred, which systems were involved, and what evidence supports the findings.

RECOMMENDATIONS

- Enable multi-factor authentication (MFA) for all official social-media and administrative accounts.
- Use strong, unique passwords and secure password-management practices.
- Restrict administrative privileges to authorized personnel and review them regularly.
- Review active sessions and revoke unknown or unused sessions.
- Enable login and security alerts and monitor unusual account activity.
- Secure recovery email accounts and phone numbers with MFA.
- Maintain appropriate account, website and network logging with suitable retention.

DIGITAL FORENSICS PROCESS FLOWCHART

Social Media Account Hacking Investigation – The 6 Phases of a Digital Forensic Investigation

PHASE	OBJECTIVE	KEY ACTIVITIES	OUTPUT / DELIVERABLE
1 IDENTIFICATION Recognize and define the	• Receive incident report • Identify compromised accounts/systems • Define investigation questions	Incident Report Scope & Plan	
2 PRESERVATION Protect evidence from	• Secure account without destroying evidence • Preserve posts/messages/logs • Isolate relevant devices where appropriate	Preservation Log Chain of Custody	
3 COLLECTION Gather relevant digital	• Account/login records • Website/server logs • Admin device images/extractions • Email security records	Collected Evidence Set Forensic Copies	
4 EXAMINATION Process collected data and	• Examine login/session history • Review posts/messages • Check account-setting changes	Examined Data Set Extracted Artifacts	
5 ANALYSIS Interpret evidence to	• Build incident timeline • Correlate account/device/IP evidence • Identify likely compromise indicators	Timeline of Events Analysis Notes	
6 REPORTING Document findings and	• Summarize incident and evidence • Present timeline and findings • State limitations	Forensic Report Evidence Summary	

KEY PRINCIPLES THROUGHOUT ALL PHASES	Integrity	Documentation	Chain of Custody	Confidentiality	Objectivity	Admissibility

DIGITAL EVIDENCE & CHAIN OF CUSTODY

The chain of custody documents the handling history of each evidence item. Because this case study is a hypothetical investigation rather than an actual evidence seizure, the record below is intentionally left blank. Investigators would complete it during real evidence handling with the actual date, time, action, handler and signature.

Evidence ID	Evidence Description	Source	Purpose / Relevance
E-01	Social-media login/session records	Official account	Identify suspicious access and timestamps
E-02	Unauthorized posts and messages	Social-media account	Establish misuse and impact
E-03	Security/recovery notifications	Institutional email	Identify takeover indicators
E-04	Administrator device image/extraction	Computer / phone	Examine local account artifacts
E-05	Website/server logs	REC infrastructure	Correlate administrative activity
E-06	Network logs	Firewall/VPN/proxy	Correlate network access
E-07	CCTV footage	Relevant premises	Correlate physical access

Sample Chain-of-Custody Record – To Be Completed During Actual Evidence Handling

Date / Time	Evidence ID	Action	Handled By	Signature
	E-01	Collected / Exported		
	E-01	Transferred to secure forensic storage		
	E-01	Examined / Accessed		
	E-01	Returned / Secured		
	E-02	Collected / Preserved		
	E-02	Examined / Accessed		

EVIDENCE ANALYSIS MATRIX

The following matrix connects each evidence source to the investigation question it can help answer. This makes the relationship between evidence, examination and analysis clear.

Evidence	Question Addressed	How It Is Correlated	Possible Limitation
Login/session records	Who/what accessed the account?	Compare IP, device, time and session data with other evidence.	IP/account may not identify a person by itself.
Posts/messages	What unauthorized activity occurred?	Compare creation timestamps with suspicious logins and reports.	Platform may limit historical metadata.
Security alerts	Was a new device or recovery action used?	Match alerts with login and account-setting changes.	Notifications may be incomplete or unavailable.
Admin device	Was the account accessed locally?	Examine browser, application and system artifacts.	Artifacts may have been deleted or overwritten.
Network logs	Which network source was involved?	Correlate IP/time with platform records.	Logs may have limited retention.
CCTV	Was physical access possible?	Compare camera timestamps with device/account events.	Camera coverage or timestamps may be incomplete.

FORENSIC REASONING PRINCIPLE

A strong forensic conclusion comes from **correlation**. For example, an unfamiliar login becomes more meaningful when its timestamp matches the creation of an unauthorized post, a security alert, and a corresponding network/device record. Likewise, a USB connection should not be treated as proof of compromise unless other evidence supports the connection.

CONCLUSION

This case study demonstrated the application of the Digital Forensics Investigation Process to a suspected social-media account compromise at REC. The incident involved unauthorized posts and suspicious money-request messages being sent from an institutional account. The reports received from parents prompted REC management to escalate the incident to the principal for forensic investigation.

By systematically following Identification, Preservation, Collection, Examination, Analysis and Reporting, the forensic team can establish a disciplined approach to investigating the incident. Identification defines the scope and possible evidence. Preservation protects fragile digital information. Collection gathers account, device, email, server, network and physical evidence. Examination extracts relevant artifacts, while Analysis correlates them to reconstruct the timeline and assess possible access sources.

The investigation also demonstrates why digital evidence must be interpreted carefully. A single IP address, account login, device or USB connection does not automatically prove who committed an incident. Reliable conclusions require corroboration from multiple evidence sources and proper documentation of the evidence-handling process.

Finally, the case highlights the importance of preventive controls such as multi-factor authentication, strong unique passwords, restricted administrator access, security alerts, logging, staff awareness and a documented incident-response procedure. These controls can reduce both the likelihood and impact of future account compromises.

OVERALL FINDING

Based on the facts supplied for the case study, REC experienced a suspected unauthorized compromise of its official digital presence. The available forensic process can be used to determine the incident timeline, affected systems, access indicators and supporting evidence. A definitive identification of the attacker or exact compromise technique would require access to the actual account, device, email, server and network evidence.

— END OF CASE STUDY —